

Project Requirement Document

DOCUMENT NUMBER

PROJECT TITLE: "Fingerprint Log IN"

CLIENT: Resource Manager





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Team Structure

- Project Manager : Daffa Aryza Pasya
- Client : Resource Management
- Project Officer : Amalia Muti'ah Khairunnisa
- Team
 1. Shadiq Arif Aryasatya
 2. Chelsea Natasja Jesslyne Sembiring
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Project Overview

1. Executive Summary

The “Fingerprint Log In” project is an initiative undertaken to enhance resource management at the OmahTi basecamp. The project aims to address the challenge of tracking member attendance by implementing a biometric authentication system based on fingerprints. The system enables members to clock in and clock out, records attendance duration, and generates reports and a leaderboard based on accumulated time.

Domain Objectives: OmahTI requires a solution to accurately monitor basecamp usage, ensuring optimal utilization of resources such as space and facilities while improving transparency in attendance management

Project Objectives:

- Develop a secure and efficient fingerprint-based authentication system to log member attendance.
- Provide real-time reporting through integration with Google Sheets for managerial purposes.
- Generate a leaderboard to visualize accumulated attendance time as a gamification feature.

Key Constraint: The absence of a budget for physical prototyping necessitates development through Wokwi simulation using an ESP32 microcontroller.

Key Deliverables:

- Simulated ESP32 firmware for fingerprint registration and authentication.
- H2 database schema for storing fingerprint templates and attendance logs.
- Server application to process data and integrate with Google Sheets.
- Technical documentation, user manual, and project report.

This project aims to deliver a reliable, cost-effective, and user-friendly solution aligned with OmahTi’s needs, emphasizing data security and operational efficiency.

2. Design and Implementation Constraints

2.1 Budget

- Constraint: No funding is available for procuring hardware such as fingerprint sensors, physical ESP32 microcontrollers, or dedicated servers.
- Impact: Development is restricted to Wokwi simulation, which may not fully replicate real-world hardware behavior.



- Mitigation: Leverage open-source platforms and free tools like Wokwi, H2 database library, and Google Sheets API to minimize costs.

2.2 Technical

- Constraint:
 - Mandatory use of the H2 database library, which may have limitations compared to alternatives like MySQL.
 - Integration with Google Sheets API, requiring authentication setup and error handling.
 - Fingerprint data format must be compatible with the simulated sensor and H2 database.
- Impact: Potential challenges in system integration and data management.
- Mitigation: Conduct thorough testing during development and refer to official documentation for H2 and Google Sheets API.

2.3 Legal and Regulatory

- Constraint: The system handles sensitive biometric data (fingerprints), classified as personal data under Indonesia's UU PDP. Requirements include user consent, data encryption, and rights to data deletion.
- Impact: The team must design the system with robust security measures and ensure legal compliance, adding complexity to development.
- Mitigation: Implement encryption for fingerprint data, document user consent processes, and restrict database access.

3. Assumptions and Dependencies

3.1 Assumptions

- The Wokwi simulation for ESP32 and fingerprint sensor accurately represents real hardware functionality for testing purposes.
- The server infrastructure (local or cloud-based) provides sufficient uptime and security for system operations.
- The Google Sheets API can handle real-time data updates without significant disruptions.

3.2 Dependencies

- External:
 - Availability of the Wokwi platform and libraries for simulating fingerprint sensors and ESP32.



- Comprehensive documentation for the H2 database library and Google Sheets API.
- Stable internet connectivity for communication between system components (ESP32, server, database, Google Sheets).
- **Internal:**
 - Availability of team members with expertise in C/C++ programming for ESP32 firmware, server development, and API integration.
 - Effective team coordination to meet project deadlines.





Project Requirements

4. Interfaces

4.1. User Interfaces

The user interface is designed to ensure an intuitive and efficient experience for OmahTi members and administrators.

- **Key Components:**

- **Fingerprint Sensor (Simulated):** Users place their finger to clock in or clock out. In Wokwi, this is represented as a user input.

- **Feedback Mechanism:** Visual indicators (e.g., simulated LED) or text messages to confirm successful or failed authentication.

- **Reporting Dashboard:** Administrators access attendance logs and the leaderboard via Google Sheets or, if feasible, a simple web interface.

- **Design Preferences:**

- **Style:** Minimalist, prioritizing clarity and speed of interaction.

- **Accessibility:** Clear instructions to ensure usability for members with limited technical experience.

- **Appendices:** User flow diagrams, wireframes for the dashboard (if applicable), and mockups will be provided as separate documents.

4.2. Software Interfaces

The system comprises interconnected software components:

- **ESP32 Firmware:**

- Written in C/C++, responsible for capturing fingerprint data, processing authentication, and sending data to the server.

- Communicates with the server via HTTP/HTTPS protocols.

- **Server Application:**

- Handles requests from the ESP32, validates data, and updates the H2 database.

- Integrates attendance logs with Google Sheets via API.

- May be implemented in Python, Java, or another language based on team expertise.

- **H2 Database:**

- Stores tables for users (ID, name, fingerprint template) and attendance logs (user ID, clock in, clock out, duration).



- Accessed by the server via JDBC or H2 API.
- **Google Sheets API:**
 - Used by the server to write log data to a designated spreadsheet in real-time.
 - Requires OAuth 2.0 authentication for secure access.
- **Communication Protocols:**
 - HTTP/HTTPS for ESP32-server and server-Google Sheets interactions.
 - JDBC for server-database communication.

4.3. Hardware Interfaces (optional)

This section describes the project's hardware components and how the application interacts with its components.

- **ESP32:** Acts as the main microcontroller, handling authentication logic and communication.
- **Fingerprint Sensor (Virtual):** Represented as an input module in Wokwi, simulating fingerprint scanning.
- **Interaction:** The virtual sensor communicates with the ESP32 via interfaces like UART or SPI, depending on the simulation library used.

5. Functional Requirements

5.1. User Managements

- FR-1: The system must allow administrators to register new users by capturing multiple fingerprint samples to create a reliable template.
- FR-2: The system must authenticate users based on their fingerprints to perform clock in or clock out actions.
- FR-3: The system must record the time of each clock in and clock out event for every user with accurate timestamps.
- FR-4: The system must calculate the attendance duration for each user (difference between clock in and clock out).
- FR-5: The system must generate a leaderboard displaying accumulated attendance time per user, sorted from highest to lowest.

5.2. Data Management

- FR-6: The system must store user data, including unique ID, name, and fingerprint template, in the H2 database.
- FR-7: The system must maintain attendance logs, including user ID, clock in time, clock out time, and duration, in the H2 database.



- FR-8: The system must synchronize attendance logs with Google Sheets for real-time or periodic reporting.

5.3. **Security**

- FR-9: The system must encrypt fingerprint templates during storage and transmission to prevent unauthorized access.
- FR-10: The system must implement authentication mechanisms for administrator access to attendance logs and the leaderboard.

6. **Use-Case Requirements**

- **UC-1: User Registration**

- Actor: Administrator.
- Description: The administrator enters user details (name, ID) and prompts the user to place their finger multiple times to create a fingerprint template. The template is stored in the H2 database.
- Failure Condition: Poor-quality fingerprint or system failure; the user is prompted to retry.

- **UC-2: Clock In**

- Actor: OmahTi Member.
- Description: The user places their finger on the sensor; the system verifies the fingerprint, logs the clock in time, and provides feedback.
- Failure Condition: Unrecognized fingerprint; the user is prompted to retry up to a set limit.

- **UC-3: Clock Out**

- Actor: OmahTi Member.
- Description: The user places their finger after a clock in; the system verifies, logs the clock out time, calculates duration, and updates the log.
- Failure Condition: No prior clock in; the system displays an error.

- **UC-4: Reporting and Leaderboard**

- Actor: Administrator.
- Description: The administrator accesses Google Sheets to view attendance logs and the leaderboard based on accumulated time.
- Failure Condition: Data synchronization failure; the administrator must check connectivity or API configuration.

7. **Non-Functional Requirements**

- **Performance:**

- Fingerprint authentication must complete within 3 seconds to ensure a seamless user experience.



- Data synchronization to Google Sheets must complete within 5 seconds per update.
- **Reliability:**
 - Fingerprint recognition accuracy must exceed 95% to minimize false rejection rate (FRR) and false acceptance rate (FAR).
 - The system must maintain a server uptime of at least 99% for reliable operation.
- **Security:**
 - Fingerprint templates must be encrypted using algorithms like AES-256 during storage and transmission.
 - Access to the database and reporting dashboard must be restricted with password or token-based authentication.
 - The system must comply with UU PDP, including providing users the right to delete their data.
- **Scalability:**
 - The system must support up to 100 users without performance degradation.
 - The H2 database must handle thousands of attendance logs efficiently.
- **Usability:**
 - The clock in/clock out process must require minimal steps (single finger touch).
 - User documentation must be clear, with step-by-step guides for administrators and members.



Project Execution

8. Project Scope

- **In Scope:**
 - Develop simulated ESP32 firmware for fingerprint registration and authentication.
 - Set up an H2 database with the required schema.
 - Create a server application for data processing and Google Sheets integration.
 - Provide technical documentation and user manuals.
- **Out of Scope:**
 - Development and testing of physical hardware.
 - Advanced features like facial recognition or other biometrics.
 - Mobile application development.

9. Project Deliverables

- Source code for simulated ESP32 firmware.
- H2 database schema and setup scripts.
- Source code for the server application.
- Integration scripts for Google Sheets API.
- User manual and technical documentation.

10. Timeframe and Milestones

No	Timeframe	Milestone
1.	16th March 2025	Topic Finalization & Requirement Gathering
2.	27th March 2025	Mechanic Design Review
3.	28th March 2025	Backend Integration Consultation
4.	30th April 2025	Code Development Discussion
5.	1st May 2025	Final Deliverables Completion (PRD, Pitch Deck, Code)



11. Challenges and Open Questions

- **Fingerprint Data Format:** Determining the appropriate format for storing fingerprint templates in the H2 database.
- **Simulation Accuracy:** Ensuring Wokwi simulation accurately reflects fingerprint sensor and ESP32 behavior.
- **Security Measures:** Implementing robust security to protect biometric data, particularly in a simulated environment.
- **Compliance:** Understanding and adhering to specific UU PDP requirements for biometric data handling.
- **Integration:** Seamlessly integrating system components, especially the server with the H2 database and Google Sheets API.



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